

PRACTICAL CRYPTOGRAPHY

Algorithms and Implementations Using C++



Edited by
Saiful Azad
Al-Sakib Khan Pathan

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AN AUERBACH BOOK

ADVANCED ENCRYPTION STANDARD

ASIF UR RAHMAN, SAEF ULLAH
MIAH, AND SAIFUL AZAD

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Keywords

Advanced Encryption Standard
Block cipher

6.1 Overview

The Advanced Encryption Standard (AES) is a renowned symmetric key algorithm that utilizes a same secret key to encrypt and decrypt a message. It overcomes the limitation of the smaller key size of the Data Encryption Standard (DES) by utilizing

Cryptography, the science of encoding and decoding information, allows people to do online banking, online trading, and make online purchases, without worrying that their personal information is being compromised. The dramatic increase of information transmitted electronically has led to an increased reliance on cryptography. This book discusses the theories and concepts behind modern cryptography and demonstrates how to develop and implement cryptographic algorithms using C++ programming language.

Written for programmers and engineers, **Practical Cryptography** explains how you can use cryptography to maintain the privacy of computer data. It describes dozens of cryptography algorithms, gives practical advice on how to implement them into cryptographic software, and shows how they can be used to solve security problems.

Covering the latest developments in practical cryptographic techniques, this book shows you how to build security into your computer applications, networks, and storage. Suitable for undergraduate and postgraduate students in cryptography, network security, and other security-related courses, this book will also help anyone involved in computer and network security who wants to learn the nuts and bolts of practical cryptography.

Features

- Discusses the theories and concepts behind modern cryptography
- Describes dozens of cryptography algorithms
- Covers recent developments in practical cryptographic techniques
- Explains how to implement the algorithms using C++ and supplies the source code



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6000 Broken Sound Parkway, NW
Suite 300, Boca Raton, FL 33487
711 Third Avenue
New York, NY 10017
2 Park Square, Milton Park
Abingdon, Oxon OX14 4RN, UK

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